The Problem with Google Wallet for Receipt Management

Limited Receipt Capture

Google Wallet primarily tracks digital payments, but struggles with capturing and managing physical paper receipts or transactions made outside the Wallet ecosystem.

Basic Categorization

While it categorizes transactions, Google Wallet offers limited depth for custom categories, making detailed expense tracking and budgeting challenging for users.

Lack of Export & Integration

Exporting transaction data for tax purposes or integrating with accounting software is often cumbersome or not directly supported, requiring manual workarounds.

No AI-Powered Insights

Unlike Smart Scan, Google Wallet lacks advanced AI and OCR technology to automatically extract data, analyze spending patterns, or identify potential savings from receipts.

Key Features of Smart Scan

AI Data Extraction

AI-powered automatic data extraction from receipts, ensuring precision and reducing the need for human intervention.

High-Accuracy OCR

Leverages advanced Optical Character Recognition (OCR) for superior data accuracy across various receipt formats.

Instant Organization

Expenses are instantly categorized and organized, providing a clear overview of your spending patterns.

Multi-Format Support

Supports JPEG, PNG, PDF, TIFF, WEBP, HEIC, and HEIF, offering flexibility for all your digital receipt needs.

Seamless Integration

Authentication using Firebase Auth(google login) and wallet sync simulated via transaction data mock from Firestore.

Flowchart: Smart Scan Workflow

User Uploads Receipt

User captures or uploads a receipt via the frontend (HTML/CSS/JS or Flutter).

Secure Cloud Storage

Receipt images are stored securely in Firebase Cloud Storage.

AI & OCR Processing

Vertex AI + Gemini 2.5 + Document AI / Vision OCR extract data (merchant, date, total, etc.).

Backend Functions

Firebase Cloud Functions (Node.js/Python) process extracted data and apply business logic.

Data Persistence

Processed financial data is stored in Firestore (NoSQL) for quick retrieval and analysis.

Wallet Sync & Display

Transaction data is synchronized (mocked via Firestore) and displayed to the user via the frontend, with Firebase Auth for Google login.

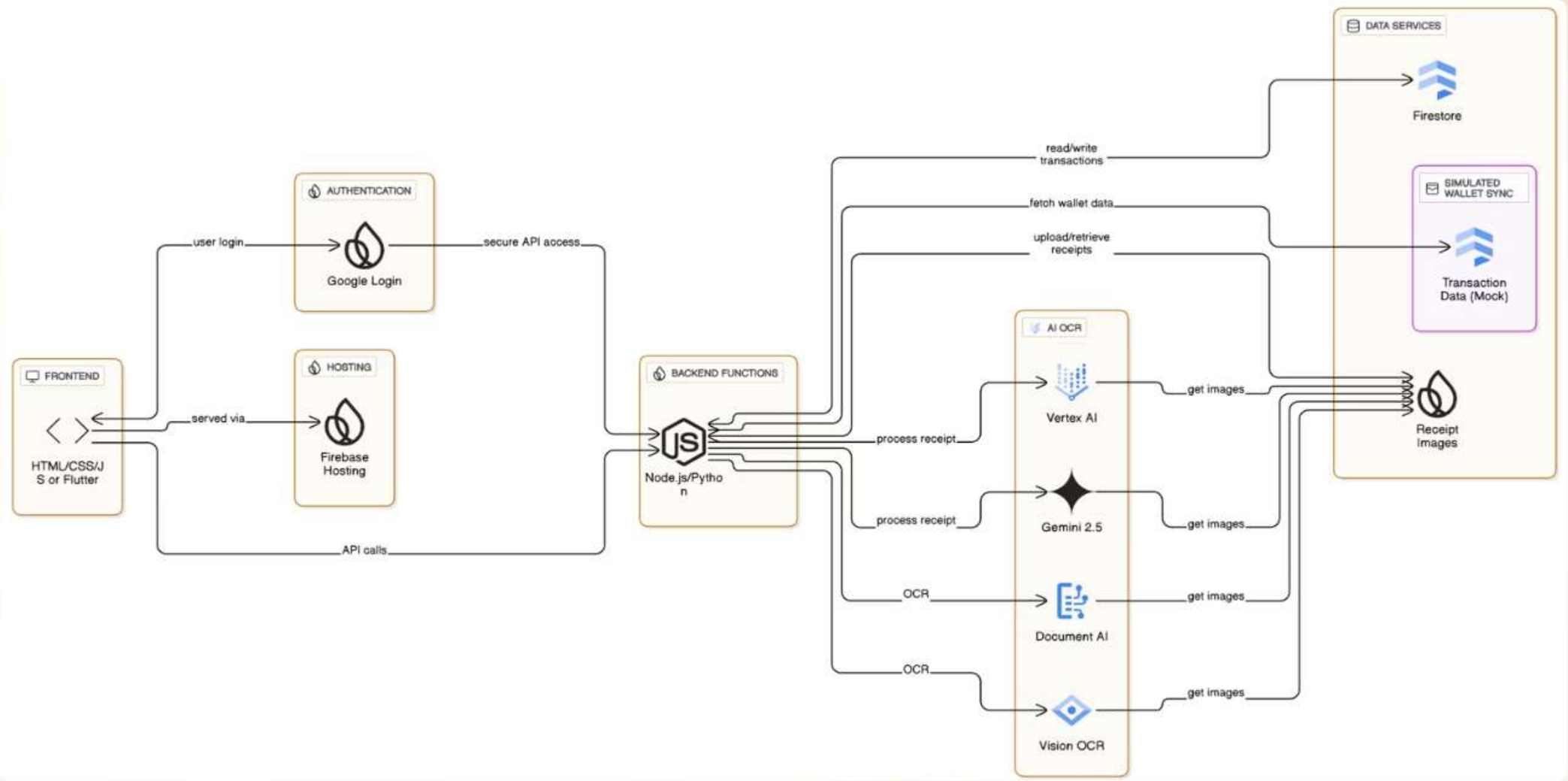
How Smart Scan Works: AI & OCR Technology

- Utilizes advanced OCR to accurately read printed text on diverse receipt types.
- AI algorithms intelligently extract critical data: merchant, date, total, VAT, and payment method.
- Processes documents rapidly in 1–2 seconds, achieving over 90% automation rate.
- Supports multi-language receipts, making it ideal for global and international use.
- Cloud-based API provides instant JSON data output for seamless integration with other applications.

Smart Scan Technology Stack

Frontend	HTML/CSS/JS or Flutter (for cross-platform)
Backend Functions	Firebase Cloud Functions (Node.js / Python)
Database	Firestore (NoSQL)
AI/OCR	Vertex AI + Gemini 2.5 + Document AI or Vision OCR
Storage	Firebase Cloud Storage (for receipt images)
Hosting	Firebase Hosting
Authentication	Firebase Auth (Google login)
Wallet Sync	(Simulated) via transaction data mock from Firestore

Tech Stack Diagram



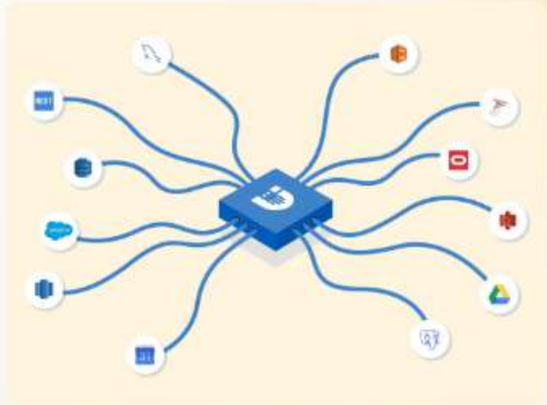
AI + Finance = Smart Control

Smart Receipt Capture & Extraction

- User uploads a paper receipt or clicks a photo
- Uses **Google Vision OCR** + **Vertex AI (Gemini)** to extract merchant, date, amount, items, and tax
- Automatically saves structured data in Firestore

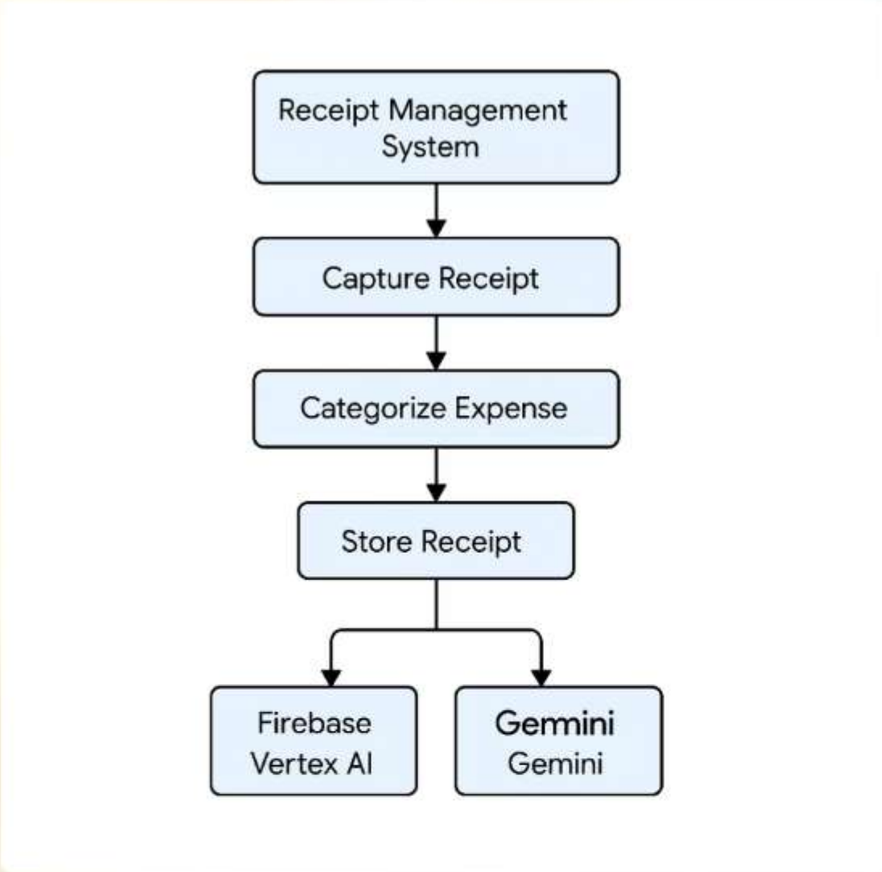
AI-Based Categorization & Monthly Insights

- Categorizes each receipt as Food, Travel, Shopping, etc. using **Gemini's function-calling**
- Generates **monthly spending summaries**: "You spent ₹9,300 on Food in June"
- Shows pie charts & trends in a clean dashboard



Unified Receipt Management

- Receipts can come from:
  - Camera or upload
  - Gmail (e-receipts)
  - Wallet transaction sync (mocked or future API)
- All stored securely in **Firebase Firestore + Cloud Storage**



🧾 1. **Captures Receipts**

- From camera 📷 (paper bills), uploads ,Gmail inbox (e.g., online orders),Google Wallet (transaction records, mocked or real API)

🧠 2. **Extracts Useful Data :**Using AI (OCR + Gemini)-

- Merchant name (e.g., Zomato, Amazon),Date and time,Amount spent,Line items (e.g., "2x Burger ₹250") etc

🏷️ 3. **Auto-Categorizes Expenses**

- Labels the receipt: **Food, Travel, Utilities, Shopping, etc.**
- Tags frequent merchants
- Adds contextual tags (e.g., "work", "personal", "reimbursable")

🔍 4. **Stores & Organizes Receipts**

- Saves receipt data + image in a structured way (Firestore + Cloud Storage)
- Allows search by date, category, amount, tags, or merchant
- Can filter or export (CSV, PDF)

📊 5. **Analyzes Spending**

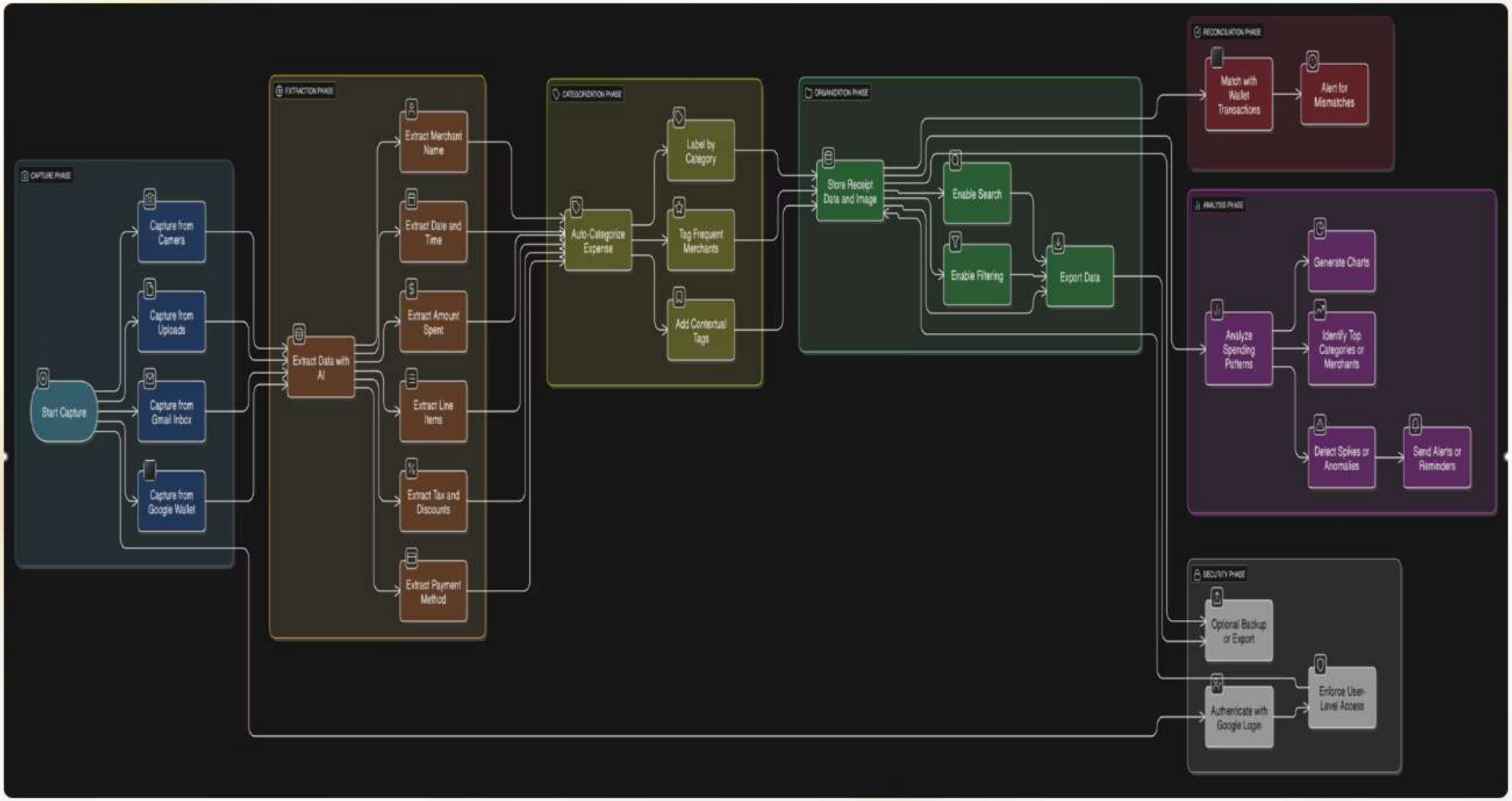
- Monthly/weekly/daily charts,Top categories or merchants
- Detects spikes or anomalies
- Sends alerts/reminders ("You spent ₹3000 more on food this month")

🤖 6. **Automates Reconciliation**

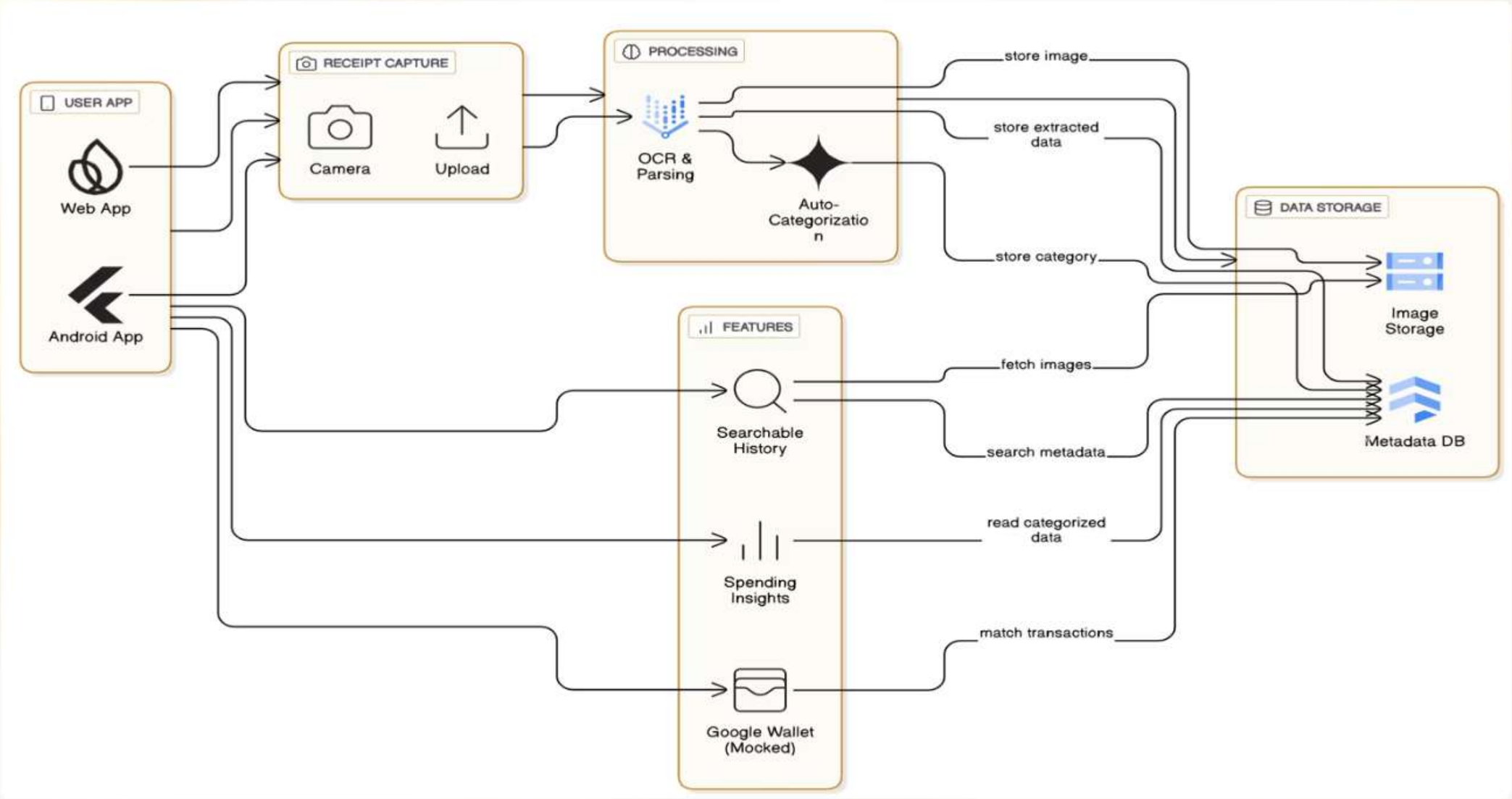
- Matches receipts with Google Wallet transactions (mocked or real)
- Alerts for mismatches (e.g., missing receipt for a ₹950 debit)

🔐 7. **Secures Your Data**

- Google Login (Firebase Auth)
- Cloud Firestore rules for user-level access
- Optional backup/export



Process Flow diagram


USER APP
Web App
Android App
RECEIPT CAPTURE
Camera
Upload
PROCESSING
OCR & Parsing
Auto-Categorizatio n
FEATURES
Searchable History
Spending Insights
Google Wallet (Mocked)
store image
store extracted data
store category
fetch images
search metadata
read categorized data
match transactions
DATA STORAGE
Image Storage
Metadata DB

Use Cases & Target Users

Small Businesses & Freelancers

Automating bookkeeping and expense tracking to reclaim valuable time and reduce administrative burden.

Finance Teams

Reducing administrative workload, minimizing manual errors, and streamlining audit processes for greater efficiency.

Traveling Professionals

Effortlessly uploading receipts via SMS or email while on the go, ensuring all expenses are captured immediately.

Contractors

Tracking project-specific expenses in real-time, simplifying client billing and financial oversight for individual projects.

Individuals

Managing personal budgets, categorizing household expenses, and organizing tax documents for simplified annual filing.

Google Wallet vs. SmartScan: A Feature Comparison

Google Wallet Shortcoming

- No native receipt storage
- Limited AI categorization
- No scan-to-save feature
- Basic spending insights
- No e-receipt synchronization
- Limited export options
- Absence of smart alerts

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Thank You

We appreciate your time and interest in Smart Scan.